

vieo

Video Streaming Platform — User Guide

Contents

- 1. [Overview](#)
- 2. [Getting Started](#)
- 3. [Managing Channels](#)
- 4. [Adding Sources](#)
- 5. [Transcoding Streams](#)
- 6. [Watching Streams](#)
- 7. [Managing Jobs](#)
- 8. [Dashboard](#)
- 9. [Troubleshooting](#)

1. Overview

vieo is a self-hosted video streaming platform that takes your media sources (files, URLs, live streams) and converts them into web-friendly HLS format for playback in any modern browser.

What can vieo do?

- **Transcode video files** — Convert MP4, MKV, and other formats to adaptive HLS streaming
- **Stream live sources** — Ingest RTMP, RTSP, and HLS streams and make them playable in the browser
- **Live preview** — Watch streams in real-time as they are being transcoded
- **Organize with channels** — Group related streams into channels for easy management
- **Monitor everything** — Real-time progress tracking, disk usage monitoring, and job management

Supported input types

Type	Description	Example
File / URL	Local files or remote HTTP URLs	<code>/path/to/video.mp4</code> , <code>https://example.com/video.mkv</code>
HLS	HLS master playlists	<code>https://example.com/stream/master.m3u8</code>
RTMP	RTMP live streams	<code>rtmp://server/live/stream_key</code>
RTSP	RTSP camera/network streams	<code>rtsp://192.168.1.100:554/stream</code>
Device	Local capture devices	<code>/dev/video0</code>

2. Getting Started

1 Open vido in your browser

Navigate to `http://localhost:8080` (or the address your administrator configured).

2 Create your first channel

Go to **Channels** in the sidebar and click **New Channel**. Give it a name (e.g., "My Streams") and an optional description.

3 Add a source

Open your channel and click **Add Source**. Select the source type and enter the URL or file path.

4 Start transcoding

Click **Start Transcoding** on your source. The job will appear on the **Jobs** page with a real-time progress bar.

5 Watch your stream

Once transcoding starts, click the **Play** button on the job to open the player. You can watch while transcoding is still in progress.

Tip: The player works while transcoding is running — you don't need to wait for it to finish. Early segments become available within seconds.

3. Managing Channels

Channels are organizational containers for your media sources. Think of them as folders or categories.

Creating a channel

1. Click **Channels** in the sidebar
2. Click **New Channel**
3. Enter a **Name** (required) — this is what appears in the UI
4. Enter a **Slug** (required) — a URL-friendly identifier (e.g., `my-streams`)
5. Enter a **Description** (optional)
6. Click **Create**

Editing a channel

1. Click **Channels** in the sidebar
2. Click the **Edit** (pencil) icon on the channel card
3. Modify the name, slug, or description

4. Click **Update**

Deleting a channel

1. Click **Channels** in the sidebar
2. Click the **Delete** (trash) icon on the channel card

Warning: Deleting a channel also deletes all sources, outputs, and jobs associated with it. This action cannot be undone.

4. Adding Sources

A source represents a media input — a file, URL, or live stream that video will transcode.

Adding a source to a channel

1. Open a channel by clicking **View** on a channel card
2. Click **Add Source**
3. Select the **Type**:
 - **File / URL** — For local files or HTTP URLs
 - **HLS Stream** — For HLS master playlists (.m3u8)
 - **RTMP** — For RTMP live streams
 - **RTSP** — For RTSP camera streams
 - **Device** — For local capture devices
4. Enter the **URL or file path**
5. Click **Add**

Note: The stream type (audio+video, video-only, audio-only) is automatically detected when you start transcoding. You don't need to specify it manually.

Supported URL formats

Type	Format	Example
HTTP file	<code>https://...</code>	<code>https://example.com/video.mp4</code>
Local file	<code>/path/to/...</code>	<code>/home/user/videos/recording.mkv</code>
HLS	<code>https://...m3u8</code>	<code>https://example.com/stream/master.m3u8</code>
RTMP	<code>rtmp://...</code>	<code>rtmp://server/live/key</code>
RTSP	<code>rtsp://...</code>	<code>rtsp://192.168.1.100/stream</code>

5. Transcoding Streams

Transcoding converts your source media into HLS format — a sequence of small `.ts` video segments and an `.m3u8` playlist file that browsers can stream.

Starting a transcode

1. Open a channel and find your source
2. Click **Start Transcoding**
3. vimeo will:
 - a. Create an output directory for the HLS files
 - b. Probe the source to determine codec, duration, and stream type
 - c. Begin transcoding with ffmpeg
4. The job appears on the **Jobs** page with a progress bar

What happens during transcoding?

- **Probing** — vimeo analyzes the source to detect video/audio streams, resolution, and duration
- **Segment creation** — ffmpeg splits the output into 4-second `.ts` segments
- **Playlist updates** — The `.m3u8` playlist is updated as each segment completes
- **Progress tracking** — Real-time progress is shown based on the output timestamp vs. total duration

Output format

Property	Value
Container	MPEG-TS (<code>.ts</code> segments)
Video codec	H.264 (libx264, fast preset, CRF 23)
Audio codec	AAC (128kbps)
Segment duration	4 seconds
Playlist	Full list (all segments)

6. Watching Streams

vimeo includes a built-in web player powered by hls.js that works in all modern browsers.

Opening the player

- **From the Jobs page:** Click the **Play** icon on any running or completed job
- **From the Dashboard:** Click on any stream thumbnail in the "Live Streams" section
- **From Channel Detail:** Click **View Stream** on a source with an active output

Live preview (during transcoding)

You can watch a stream while it's still being transcoded. The player automatically switches to live mode when viewing a running job:

- The player buffers a few segments ahead and plays from the most recent content
- If a segment isn't written yet, the player retries automatically
- Once transcoding completes, the player switches to on-demand mode for full seeking

Tip: For the best live preview experience, wait at least 8-12 seconds after starting a transcode before opening the player. This allows a few segments to be written.

Player controls

Control	Description
Play / Pause	Toggle playback
Seek bar	Jump to any point in the stream (available after transcoding completes)
Volume	Adjust audio volume
Fullscreen	Expand to fill the screen
Back	Return to the previous page

Note: The player page is independent from the rest of the UI — navigating away from it (e.g., to check job progress) won't interrupt playback if you use the browser's back button.

7. Managing Jobs

A job represents a single transcoding task — converting one source into HLS output.

Job statuses

Status	Description	Actions available
Pending	Waiting for a transcoding slot	Stop, Delete
Running	Currently transcoding	Pause, Stop, Play, Delete
Paused	Temporarily suspended (disk low or manual)	Stop, Retry, Delete
Completed	Transcoding finished successfully	Play, Retry, Delete
Failed	Transcoding error occurred	Retry, Delete
Stopped	Manually stopped by user	Retry, Delete

Job actions

- **Pause** — Temporarily suspends transcoding. The job can be resumed later from where it left off.
- **Stop** — Stops the job permanently. The output is kept but marked as stopped.
- **Retry** — Restarts the job from the beginning. Existing output files are cleaned before restarting.
- **Delete** — Removes the job record from the database.

Filtering jobs

Use the status dropdown at the top of the Jobs page to filter by status (e.g., show only "Running" jobs).

Progress tracking

Running and paused jobs display a progress bar showing the current transcoding progress as a percentage. Progress is updated in real-time via WebSocket.

8. Dashboard

The Dashboard provides an overview of your streaming platform at a glance.

Statistics cards

Card	Description
Channels	Total number of channels
Sources	Total number of media sources
Running Jobs	Currently active transcoding jobs
Completed	Successfully completed jobs
Failed	Jobs that encountered errors
Paused	Jobs paused due to disk issues or manual action

Live Streams

Shows thumbnails for all running and completed streams. Click any thumbnail to open the player.

Recent Jobs

A list of the 5 most recent jobs with their status badges.

Storage Banner

If disk usage exceeds the warning threshold, a banner appears at the top of every page showing the current usage and free space.

9. Troubleshooting

Stream won't play

- **Check the job status** — If it shows "Failed", the error message will explain why
- **Wait for segments** — If transcoding just started, wait 10-15 seconds for the first segments to be written
- **Check the URL** — Make sure the source URL is accessible from the server
- **Browser compatibility** — The player works in Chrome, Firefox, Safari, and Edge

Transcoding is slow

- **CPU-bound** — Video transcoding is CPU-intensive. Faster CPUs produce faster results
- **Source format** — Some formats (e.g., ProRes, RAW) require more processing than others

- **Resolution** — Higher resolution sources take longer to transcode

Disk usage warnings

- **Warning (90%)** — Running jobs are paused automatically to prevent disk filling up
- **Critical (95%)** — All jobs are force-stopped
- **Recovery** — Jobs resume automatically when disk usage drops below the warning threshold
- **Manual cleanup** — Delete old jobs and their output to free space

Job stuck in "Pending"

- **Max jobs reached** — vimeo limits concurrent transcoding jobs (default: 3). Wait for another job to finish
- **Resume on restart** — Pending/running/paused jobs automatically resume when the server restarts

Source probe fails

- **Network issues** — The server must be able to reach the source URL
- **Timeout** — Remote HLS streams may take longer to probe (up to 60 seconds)
- **Format support** — Ensure ffmpeg supports the input format